Tuning report: tropical / ConvexHull

2026-10-22T19:17:19

1 Decision

Approving freezes `fishr.lambda` = 10 into every cell of the grid (frozen hash `bf951b8edf5a`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 10.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: fishr.lambda=10 is the smallest value searched: the optimum may lie beyond it, so widen the space
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): `{'optim': {'epochs': 20}}`. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	fishr.lambda	runs	div.	mean	95% CI	SD	range	
1	10	12/12	0	0.2548	[0.195, 0.31]	0.108	[0.111, 0.419]	**selected**
2	100	12/12	0	0.2581	[0.201, 0.315]	0.105	[0.121, 0.421]	
3	1000	12/12	0	0.2814	[0.24, 0.325]	0.0808	[0.19, 0.418]	
4	1e+04	12/12	0	0.3386	[0.319, 0.359]	0.0376	[0.275, 0.418]	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=100`	12	0.003362	[0.00151, 0.00541]	7	5	0	0.7744
`fishr.lambda=1000`	12	0.02662	[0.0106, 0.0432]	9	3	0	0.1460
`fishr.lambda=10000`	12	0.08378	[0.0391, 0.13]	11	1	0	0.0063

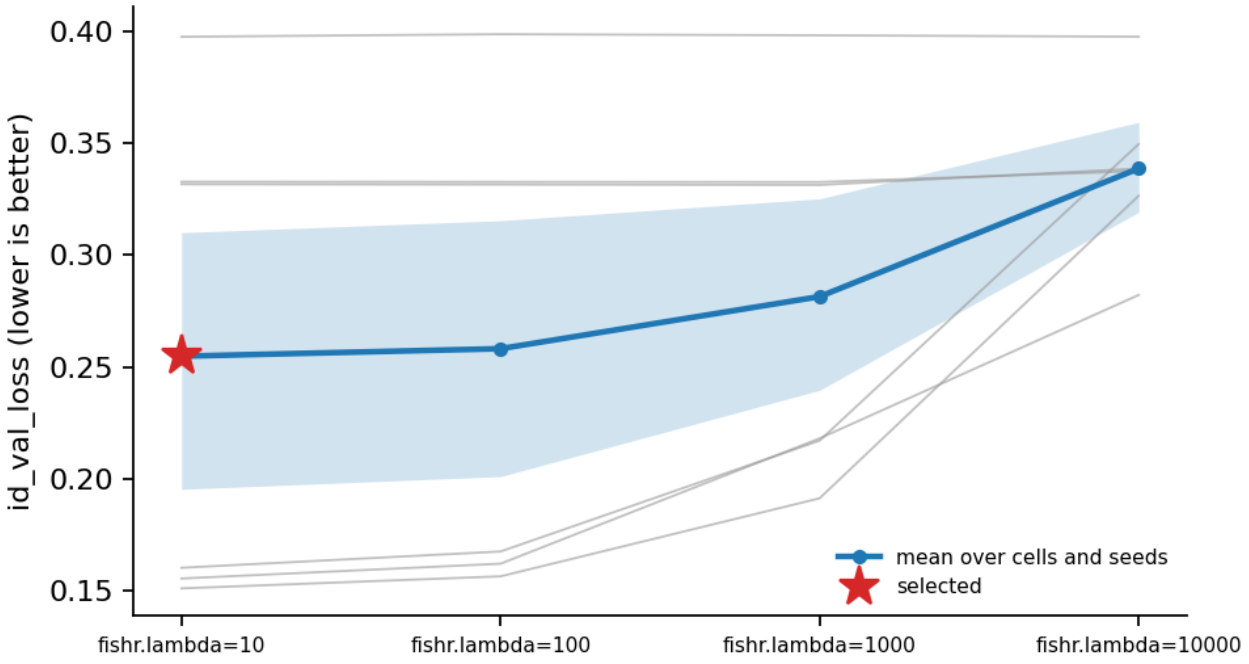


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

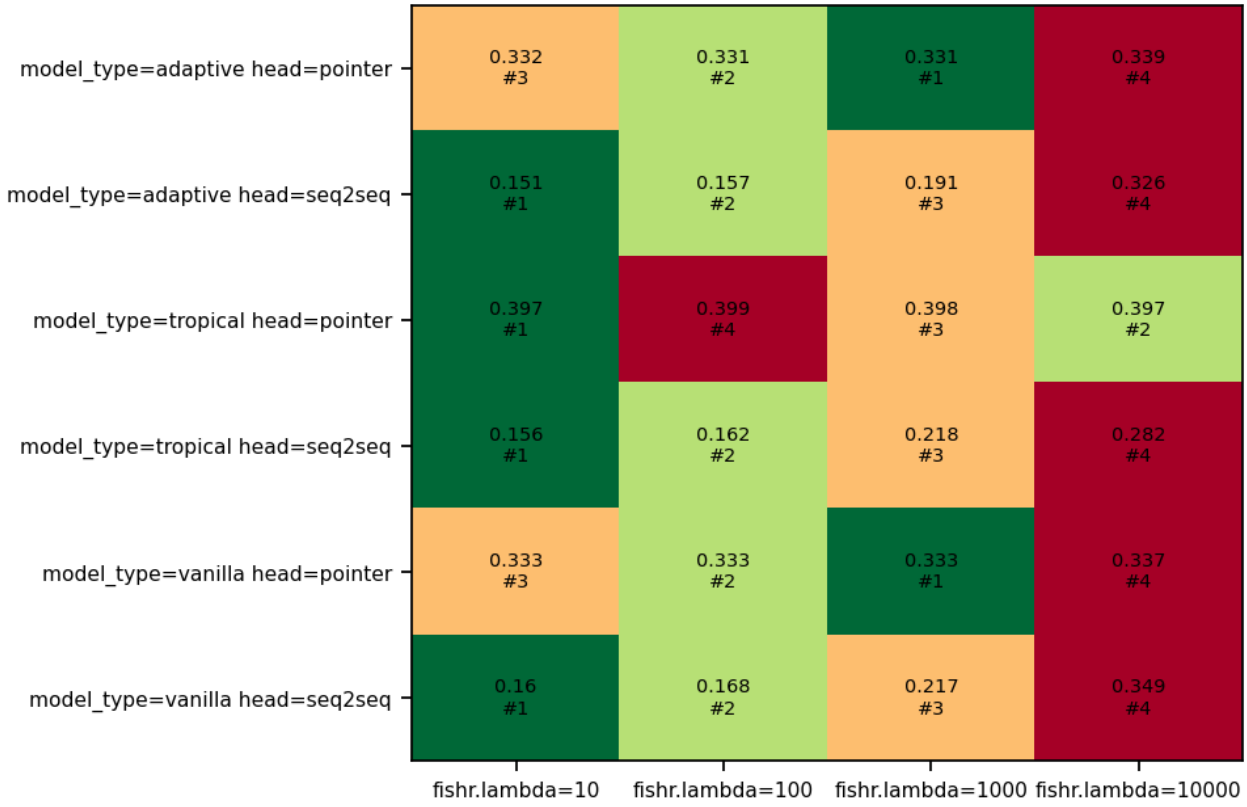


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=1000
`model_type=adaptive head=seq2seq`	fishr.lambda=10
`model_type=tropical head=pointer`	fishr.lambda=10
`model_type=tropical head=seq2seq`	fishr.lambda=10
`model_type=vanilla head=pointer`	fishr.lambda=1000
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.943	0.602	0.00118	19

Table 4: Mean of every other in-distribution metric per candidate.

candidate	id_f1	id_exact	fishr_penalty_final	best_epoch
fishr.lambda=100	0.942	0.595	4.68×10^{-4}	19
fishr.lambda=1000	0.935	0.552	1.06×10^{-4}	19
fishr.lambda=10000	0.918	0.461	3.98×10^{-5}	19

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

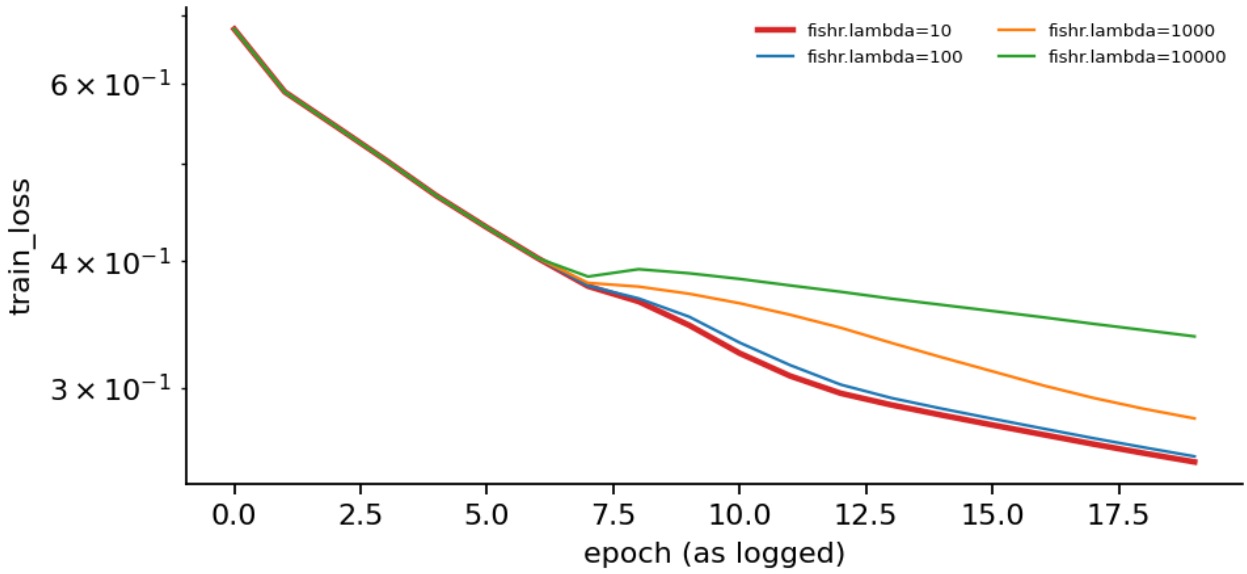


Figure 3: `train_loss` per candidate.

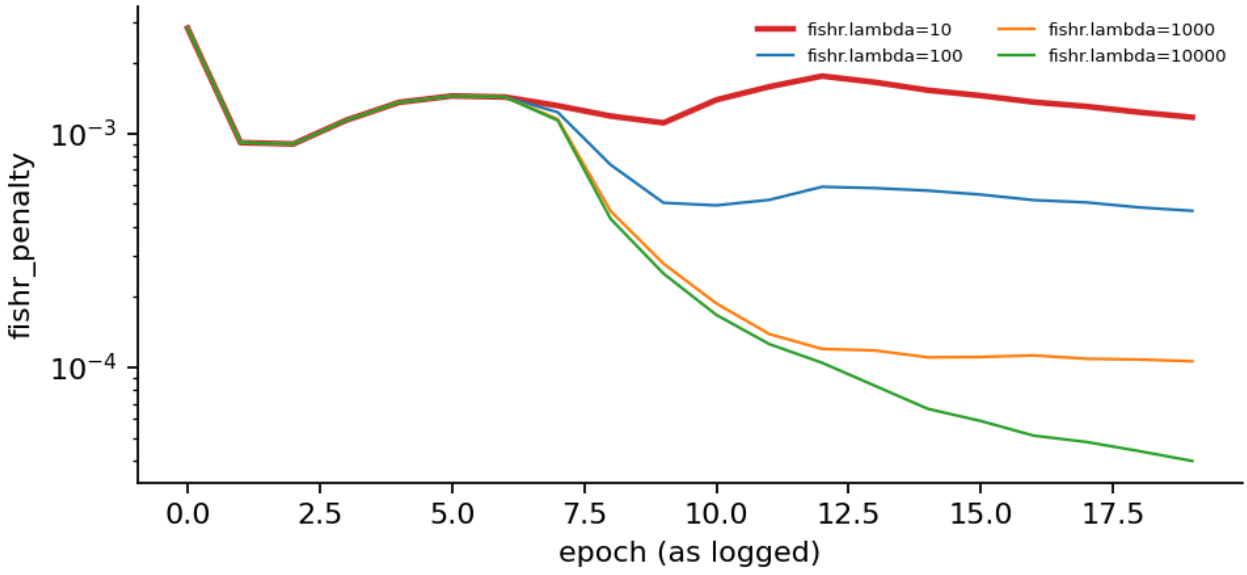


Figure 4: `fishr_penalty` per candidate.

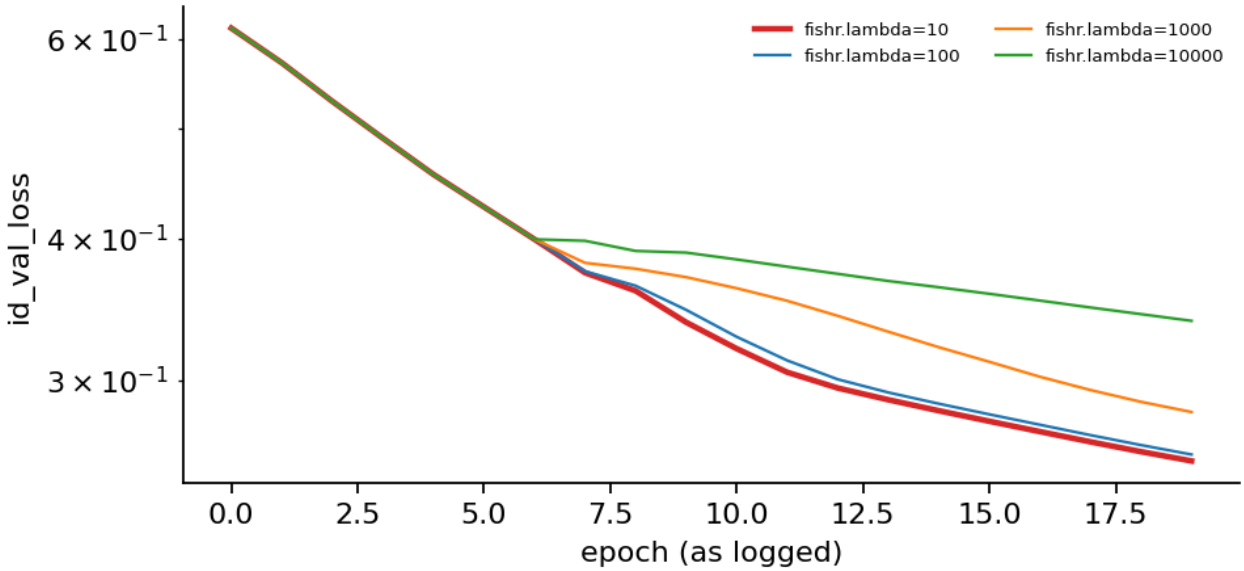


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	`6799d783f518`	12	3.62e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=100	`bc3fd56dfb98`	12	3.61e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=1000	`0157413f7779`	12	3.6e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=10000	`8a16f44b5685`	12	3.59e+03	`e5ccce8034`	2.8.0+cu128

Store: `campaigns/tropical/ConvexHull/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile ConvexHull`
Approve: `expkit approve --config configs/tropical.yaml --profile ConvexHull`

The base the grid would run with.

```
task:
  name: ConvexHullDataset
  value_range:
  - 0
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  lambda: 10.0
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16	ood_len_f1_L64	ood_len_exact_L64
fishr.lambda=10	0.657	0.124	0.789	0.132	0.823	0.192	0.904	0.256	0.853	0.112	0.482	0.00513
fishr.lambda=100	0.653	0.116	0.786	0.13	0.823	0.192	0.901	0.241	0.85	0.103	0.478	0.00321
fishr.lambda=1000	0.623	0.0822	0.79	0.132	0.822	0.187	0.885	0.184	0.832	0.0625	0.442	0
fishr.lambda=10000	0.577	0.0582	0.798	0.143	0.823	0.184	0.855	0.132	0.794	0.043	0.398	0